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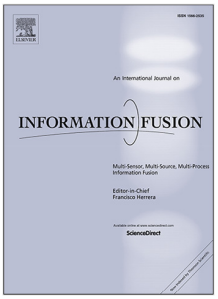

A comprehensive survey of research towards AI-enabled unmanned aerial systems in pre-, active-, and post-wildfire management

Sayed Pedram Haeri Boroujeni, Abolfazl Razi, Sahand Khoshdel, Fatemeh Afghah, Janice L. Coen, Leo O'Neill, Peter Fule, Adam Watts, Nick-Marios T. Kokolakis, Kyriakos G. Vamvoudakis

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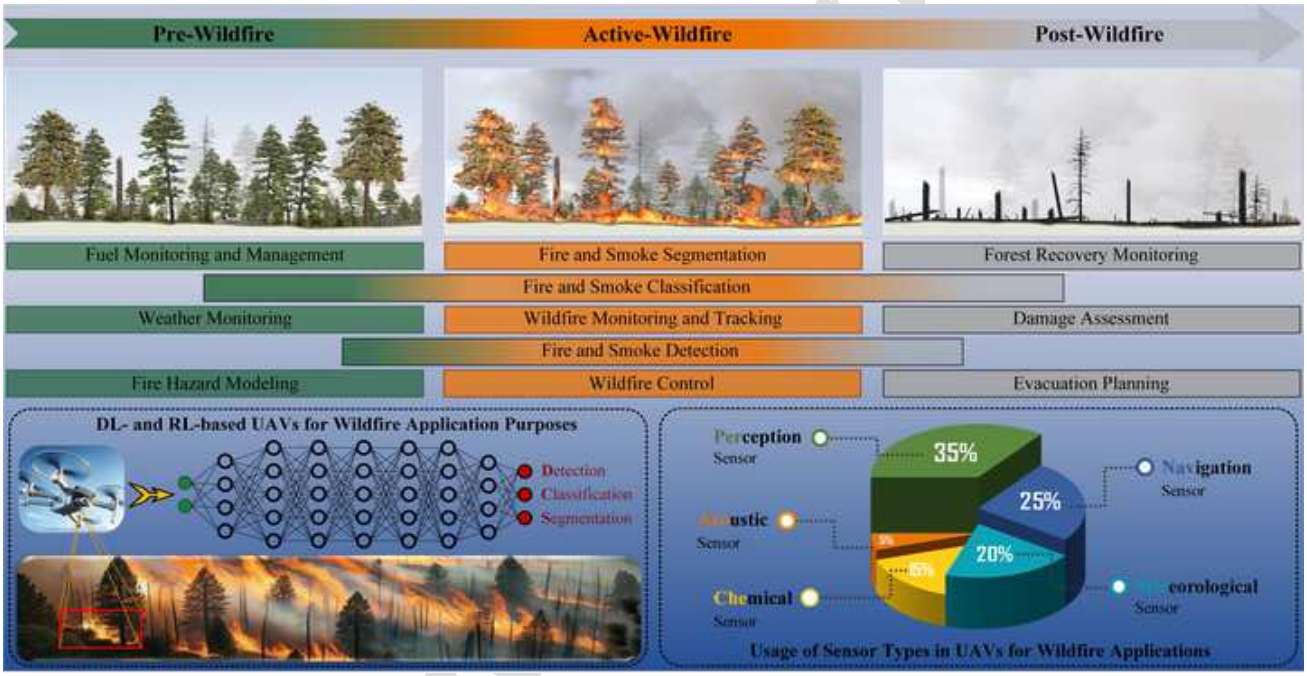

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A comprehensive survey of research towards AI-enabled unmanned aerial systems in pre-, active-, and post-wildfire management[1]

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Abstract

Wildfires have emerged as one of the most destructive natural disasters worldwide, causing catastrophic losses. These losses have underscored the urgent need to improve public knowledge and advance existing techniques in wildfire management. Recently, the use of Artificial Intelligence (AI) in wildfires, propelled by the integration of Unmanned Aerial Vehicles (UAVs) and deep learning models, has created an unprecedented momentum to implement and develop more e□ective wildfire management. Although existing survey papers have explored learning-based approaches in wildfire, drone use in disaster management, and wildfire risk assessment, a comprehensive review emphasizing the application of AI-enabled UAV systems and investigating the role of learning-based methods throughout the overall workflow of multi-stage wildfire management, including pre-fire (e.g., vision-based vegetation fuel measurement), active-fire (e.g., fire growth modeling), and post-fire tasks (e.g., evacuation planning) is notably lacking. This survey synthesizes and integrates state-of-the-science reviews and research at the nexus of wildfire observations and modeling, AI, and UAVs - topics at the forefront of advances in wildfire management, elucidating the role of AI in performing monitoring and actuation tasks from pre-fire, through the active-fire stage, to post-fire management. To this aim, we provide an extensive analysis of the existing remote sensing systems with a particular focus on the UAV advancements, device specifications, and sensor technologies relevant to wildfire management. We also examine the pre-fire and post-fire management approaches, including fuel monitoring, prevention strategies, as well as evacuation planning, damage assessment, and operation strategies. Additionally, we review and summarize a wide range of computer vision techniques in active-fire management, with an emphasis on Machine Learning (ML), Reinforcement Learning (RL), and Deep Learning (DL) algorithms for wildfire classification, segmentation, detection, and monitoring tasks. Ultimately, we underscore the substantial advancement in wildfire modeling through the integration of cutting-edge AI techniques and UAV-based data, providing novel insights and enhanced predictive capabilities to understand dynamic wildfire behavior.

Keywords: Wildfire management, Artificial intelligence (AI), Unmanned aerial vehicle (UAV),

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Machine Learning, Deep learning (DL), Reinforcement learning (RL), Computer vision.

1. Introduction

Over the past few decades, although the frequency of natural disasters has slightly decreased across the globe, their impacts have dramatically increased. They have caused extensive damage to the natural environment as well as causing severe harm to the global economy and human lives [1, 2]. These disasters can be caused by geological forces, such as earthquakes and volcanic eruptions, or by weather and climate-related events, such as wildfires, hurricanes, and floods. Their consequences pose a substantial threat not only to developing nations but also to technologically advanced developed nations. Additionally, natural disasters can have major long-term impacts such as population displacement or economic instability [3, 4]. In general, disasters are classified into two primary categories, natural and technological, and each of these categories is further divided into various subgroups. Technological disasters arise because of human-made hazards such as industrial accidents and transportation accidents, while natural disasters are caused by environmental phenomena such as geophysical forces and climatological causes, etc. Figure 1 provides a comprehensive insight into the overview of natural disasters.

Figure 1: The overview of the natural disasters and hazards.

The National Centers for Environmental Information (NCEI), as reported by the National Oceanic and Atmospheric Administration (NOAA), documented approximately 390 natural disasters worldwide in 2022. Figure 2 reveals the statistical analysis of natural hazards for five continents, including America, Asia, Africa, Europe, and Oceania in 2022. Regarding the frequency of major natural disasters, Asia recorded the highest number of disasters with 137 events, followed by the Americas with 118 events. Africa was impacted by 79 disasters, while Europe had 43 disasters, and Oceania had the lowest number of disasters with 10 events. The high occurrence of disasters in the Americas and Asia continents can be attributed to factors such as geographical locations and susceptibility to extreme weather events. In terms of the economic damages caused by natural disasters, Americas stands at the top of the list with 69.6% of total losses, which caused almost 150 billion US$ damages. Asia comes second on the list with losses of 48.7 billion US$ dollars and accounted for roughly 22% of the total losses. Africa and Oceania both su□ered 8.6 billion dollars, each accounting for 3.8% of the total losses, Europe had the smallest percentage of economic damages, with only 0.1% of the total losses.